

Diag

Time Allowed: 3 Hours

Maximum Marks: 80

General Instructions:

1. This question paper contains 40 questions. All questions are compulsory.
2. Questions are grouped into sections A through E.
3. Use of calculators is NOT permitted.

Section A: Multiple Choice & Assertion-Reason (1 mark each)

1. To draw a pair of tangents to a circle which are inclined to each other at an angle of 60° , it is required to draw tangents at the endpoints of those two radii of the circle, the angle between which is:
(A) (A) 60°
(B) (B) 90°
(C) (C) 120°
(D) (D) 180°
2. If a point P is 10 cm away from the centre of a circle of radius 6 cm, the length of the tangent drawn from P to the circle is:
(A) (A) 7 cm
(B) (B) 8 cm
(C) (C) 9 cm
(D) (D) 10 cm
3. How many tangents can be drawn from an external point to a circle?
(A) (A) 1
(B) (B) 2
(C) (C) 0
(D) (D) Infinite
4. To divide a line segment AB in the ratio 3 : 4, we draw a ray AX such that $\angle BAX$ is acute. Then we mark points on AX at equal distances. The minimum number of points to be marked is:
(A) (A) 3
(B) (B) 4
(C) (C) 7
(D) (D) 12

5. The angle between a tangent to a circle and the radius passing through the point of contact is:
- (A) (A) 45°
 - (B) (B) 60°
 - (C) (C) 90°
 - (D) (D) 180°
6. If the radius of a sphere is 7 cm, then its surface area is:
- (A) (A) 308 cm^2
 - (B) (B) 440 cm^2
 - (C) (C) 616 cm^2
 - (D) (D) 1232 cm^2
7. The volume of a right circular cylinder with radius 3 cm and height 7 cm is:
- (A) (A) 154 cm^3
 - (B) (B) 198 cm^3
 - (C) (C) 264 cm^3
 - (D) (D) 396 cm^3
8. Assertion (A): The length of the tangent drawn from an external point to a circle is always equal to the radius of the circle. Reason (R): The tangent at any point of a circle is perpendicular to the radius through the point of contact.
- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
 - (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
 - (C) (C) Assertion (A) is true but Reason (R) is false
 - (D) (D) Assertion (A) is false but Reason (R) is true
9. Assertion (A): To construct a pair of tangents to a circle from an external point, the angle between the tangents is supplementary to the angle subtended by the line segment joining the points of contact at the center. Reason (R): The sum of opposite angles in a cyclic quadrilateral is 180° .
- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
 - (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
 - (C) (C) Assertion (A) is true but Reason (R) is false
 - (D) (D) Assertion (A) is false but Reason (R) is true

10. Assertion (A): It is impossible to construct a tangent to a circle from a point lying inside the circle. Reason (R): A tangent touches the circle at only one point.
- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
- (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
- (C) (C) Assertion (A) is true but Reason (R) is false
- (D) (D) Assertion (A) is false but Reason (R) is true
11. Assertion (A): The number of tangents that can be drawn from a point on the circle to the circle is exactly two. Reason (R): A point on the circle lies on the circumference.
- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
- (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
- (C) (C) Assertion (A) is true but Reason (R) is false
- (D) (D) Assertion (A) is false but Reason (R) is true
12. Assertion (A): When constructing tangents to a circle from an external point P , we bisect the line segment OP where O is the center. Reason (R): The perpendicular bisector of OP helps in finding the midpoint which acts as the center of a circle passing through O and P .
- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
- (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
- (C) (C) Assertion (A) is true but Reason (R) is false
- (D) (D) Assertion (A) is false but Reason (R) is true

Section B: Very Short Answer (2 marks each)

- If $\tan A = \frac{4}{3}$, find the value of $\sin A + \cos A$.
- Evaluate: $\frac{2 \tan^2 45^\circ + \cos^2 30^\circ - \sin^2 60^\circ}{1}$
- If $\sin(A - B) = \frac{1}{2}$ and $\cos(A + B) = \frac{1}{2}$, where $0^\circ < A + B \leq 90^\circ$ and $A > B$, find A and B .
- Express $\cos A$ in terms of $\tan A$.
- Evaluate $\frac{\tan 65^\circ}{\cot 25^\circ}$.

6. Draw a circle of radius 3 cm. Take two points P and Q on one of its extended diameter each at a distance of 7 cm from its centre. Draw tangents to the circle from these two points P and Q .
7. Construct a tangent to a circle of radius 4 cm from a point on the concentric circle of radius 6 cm and measure its length.
8. Draw a line segment AB of length 8 cm. Taking A as centre, draw a circle of radius 4 cm and taking B as centre, draw another circle of radius 3 cm. Construct tangents to each circle from the centre of the other circle.
9. Draw a circle of radius 3.5 cm. Construct a pair of tangents to this circle which are inclined to each other at an angle of 60° .
10. Draw a circle of radius 5 cm. Mark a point P at a distance of 8 cm from the centre. Construct the two tangents from P to the circle.

Section C: Short Answer (3 marks each)

1. If $\tan(A + B) = \sqrt{3}$ and $\tan(A - B) = \frac{1}{\sqrt{3}}$, where $0^\circ < A + B \leq 90^\circ$ and $A > B$, find the values of A and B .
2. If $3 \cot \theta = 2$, evaluate the value of $\frac{4 \sin \theta - 3 \cos \theta}{2 \sin \theta + 6 \cos \theta}$.
3. Prove that $(\sin A + \csc A)^2 + (\cos A + \sec A)^2 = 7 + \tan^2 A + \cot^2 A$.
4. If $\sin \theta + \sin^2 \theta = 1$, prove that $\cos^2 \theta + \cos^4 \theta = 1$.
5. Evaluate: $\frac{\cos 45^\circ}{\sec 30^\circ + \csc 30^\circ}$.
6. Draw a circle of radius 3 cm. Take two points P and Q on one of its extended diameters each at a distance of 7 cm from its centre. Draw tangents to the circle from these two points P and Q .
7. Construct a tangent to a circle of radius 4 cm from a point on the concentric circle of radius 6 cm and measure its length.
8. Draw a line segment AB of length 8 cm. Taking A as center, draw a circle of radius 4 cm and taking B as center, draw another circle of radius 3 cm. Construct tangents to each circle from the center of the other circle.
9. Draw a circle of radius 3.5 cm. Construct a pair of tangents to this circle which are inclined to each other at an angle of 60° .
10. Construct a triangle with sides 5 cm, 6 cm and 7 cm. Then construct a circle touching the side of 6 cm at its midpoint and passing through the other vertex.

Section D: Long Answer (4-5 marks each)

1. Prove that $\frac{\cos A - \sin A + 1}{\cos A + \sin A - 1} = \csc A + \cot A$, using the identity $\csc^2 A = 1 + \cot^2 A$.

2. If $\sin \theta + \cos \theta = \sqrt{3}$, then prove that $\tan \theta + \cot \theta = 1$.
3. Evaluate: $\frac{5 \cos^2 60^\circ + 4 \sec^2 30^\circ - \tan^2 45^\circ}{\sin^2 30^\circ + \cos^2 30^\circ}$
4. If $\sec \theta = x + \frac{1}{4x}$, prove that $\sec \theta + \tan \theta = 2x$ or $\frac{1}{2x}$.
5. Draw a circle of radius 3 cm. Take two points P and Q on one of its extended diameters each at a distance of 7 cm from its centre. Draw tangents to the circle from these two points P and Q .
6. Construct a triangle ABC with side $BC = 6$ cm, $AB = 5$ cm and $\angle ABC = 60^\circ$. Then construct a triangle whose sides are $\frac{3}{4}$ of the corresponding sides of the triangle ABC . Finally, construct a tangent to the circumcircle of triangle ABC at vertex A .
7. Draw a pair of tangents to a circle of radius 4 cm, which are inclined to each other at an angle of 60° . Justify the construction.
8. Draw two concentric circles of radii 3 cm and 5 cm. Construct a tangent to the smaller circle from a point on the larger circle and measure its length. Verify the measurement by calculation.